

Saurabh Joshi, PhD

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Professional Experience

• Director of Engineering and Innovations — Oct 2024–Present

Supra, Hyderabad, India (Remote).

- Steer the technical vision, product roadmap, and architecture for Supra's core oracle (DORA), achieving sub-second latency (600–900ms).
- Design and architect Supra Native Automation to guarantee zero-block delay, tested to scale up to 50,000 automation tasks executing per block.
- Scale and lead cross-functional smart contract engineering teams through active recruitment, mentorship, and technical direction across AutoFi, DORA, Automation, and Indices.
- Formulate design specifications and architectural blueprints for next-generation initiatives like SupraLiquid/Microchain and Supra EVM.
- Direct internal security operations, leading smart contract audits for mission-critical protocols (HyperNova, DORA, AutoFi) to ensure a robust security posture.
- Spearhead language-level innovation by driving Supra-specific custom adaptations and optimizations for the Move programming language.
- Serve as a trusted member of the multisig governing authority for the Supra Foundation, managing protocol-level security and upgrades.

• Principal Engineer — May 2022–Sep 2024

Supra, Hyderabad, India (Remote).

- Partnered with leadership to establish Supra's technological roadmap and align engineering execution with long-term protocol strategy.
- Pioneered the conceptualization, mathematical design, and production delivery of the novel DORA oracle protocol, publishing a peer-reviewed research paper and securing a US patent.
- Initiated the development of the native blockchain automation service from initial feasibility research to early-stage implementation.
- Led a team of smart contract developers providing oracle and VRF integrations across multiple VM environments (EVM, Move VM, Radix, CosmWasm).
- Architected application-layer frameworks in the Move language and launched the project's formal verification and contract specification strategy.
- Coordinated and supervised the engineering team, providing hands-on technical guidance in smart contract development, consensus, and State Machine Replication (SMR).

• Staff Engineer, Product Security — Oct 2021–May 2022

Qualcomm, Hyderabad, India.

- Led and coordinated a Security Validation team of 12 engineers to design and develop proof-of-concept exploits for vulnerabilities across the mobile software stack.
- Managed a specialized team of 3 engineers conducting security validation and hardware-level countermeasure testing for mobile chipsets.
- Directed overall people management, performance evaluation, and professional development for the combined engineering team.

- Cultivated strategic university relations across India to foster academic collaborations and build a talent pipeline for the Qualcomm Security Group.
- **Assistant Professor Grade I — May 2016–Sep 2021**
Department of CSE, IIT Hyderabad, India. Tools developed: [GPURepair](#), [Open-WBO-Inc](#), [Pinaka](#), Shesha, GoProver, LLVM2GOTO
- **Assistant Professor (Grade I) — Sep 2015–May 2016**
Department of CSE, IIT Guwahati, India.
- **Post Doctoral Research Fellow — Aug 2012–Aug 2015**
Department of Computer Science, University of Oxford, UK. Tools developed: [cbmc-repair](#), [2LS](#), [Open-WBO](#), [Walcyrie](#)[^][Approved artefact for PPOPP 2016], VerifOx
- **Research Intern — May 2010–Sep 2010, Feb 2011–Apr 2011**
Rigorous Software Engineering group, Microsoft Research, India. Tools developed: Cbugs, AtomicInf
- **Project Trainee — Aug 2006–Jul 2007**
Software Engineering group, IBM India Research Lab, New Delhi, India. Tools improved: SAFE

Patents

- **“Finding Bugs with Low False Alarms and Under-specified Harness”** *Saurabh Joshi*, Shuvendu Lahiri and Akash Lal US patent number: 8793664. Patent Granted.
- **“Distributed Oracle Agreement System and Method (DORA)”**, *Saurabh Joshi*, Aniket Kate, Joshua Tobkin and David Yang, US Patent App. No: PCT/US/36923

Research Articles

- Several peer-reviewed research articles in top-tier conferences and journals in the areas of program analysis, formal verification, constraint solving and optimization, and blockchain. For details, please refer to the Publications section on my homepage: [\[Publications\]](#)

Skills

Technical Skills

- **Programming Languages:** Move C++, C, Python, Rust, Solidity, Java, C#
- **Verification Framework:** CPROVER/CBMC, Boogie
- **Compilation Framework:** LLVM
- **Scripting Languages:** bash
- **Build System:** (M)ake, CMake, Cargo
- **Cloud:** Google Cloud Compute Engines
- **Documentation:** Markdown, LaTeX, Typst
- **Version Control:** git, svn
- **Development Tools:** GitHub, Eclipse, Vim, VSCode, GitHub Actions, GDB
- **Linux System Administration**
- **AI:** Claude/Copilot/Gemini for AI assisted software engineering, Custom skills for AI agents

Soft Skills

- Excellent communication skills, both written and verbal
- Strong leadership and team management skills
- Ability to work effectively in a remote and distributed team environment
- Strong problem-solving and analytical skills
- Ability to mentor and guide junior team members

- Experience in collaborating with cross-functional teams and stakeholders
- Ability to adapt to new technologies and tools quickly

Education

- **Doctor of Philosophy**, Computer Science — Jul 2007–Feb 2013 IIT Kanpur.
 - Thesis submitted in July 2012
- **Master of Technology** (with Research Assistantship), Computer Science — July 2003–July 2006 IIT Bombay.
- **Bachelor of Engineering**, Information Technology — Sep 1999–Jul 2003 GCET, Sardar Patel University.